

TECHNICAL DOCUMENT

LTspice Simulation Methodology

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LTspice Simulation Methodology

Common setup

- Simulator: LTspice 26.0.2 for Windows.
- Device model: PTM 65-nm bulk CMOS beta model released 22 February 2006.
- Nominal condition: 1.2 V, 27 C, 125 MHz, 10 fF per observed output.
- NMOS: L=65 nm, W=120 nm.
- PMOS: L=65 nm, W=240 nm.
- Input edge time: 20 ps.
- XOR transient maximum timestep: 2 ps.
- Encoder transient maximum timestep: 20 ps.
- Power: average of $-V(VDD)*I(VSUP)$ over the documented steady-state window.
- Delay: 50%-to-50%; $tpd=(tpLH+tpHL)/2$.
- Extended encoder delay endpoint: overall parity P0.

Sweeps

The experiment set uses independent one-factor sweeps around the nominal condition: VDD = 0.6, 0.8, 1.0, 1.2 V; temperature = -20, 27, 85 C; load = 1, 5, 10, 20 fF; frequency = 25, 125, 200 MHz. A redundant full Cartesian product is intentionally avoided.

Functional criteria

For the automated operating map, a measured point is considered functionally acceptable when V_{OH} is at least 0.8 VDD, V_{OL} is at most 0.2 VDD, and propagation delay is less than half the input period. This is an explicit study criterion, not a claim that it is a foundry-qualified noise-margin specification.

Traceability

Every generated netlist and log has an experiment ID. The manifest is ``data/experiments.json``; raw LTspice logs stay next to their circuits; parsed results are in ``data/processed/results.csv``; web data are in ``thesis/data/results.json``.

Measurement Definitions

- **Average power:** time average of positive power drawn from VSUP, calculated as $AVG(-V(VDD)*I(VSUP))$.
- **tpLH / tpHL:** elapsed time from the 50% input crossing to the corresponding 50% output crossing.
- **Propagation delay:** arithmetic mean of tpLH and tpHL.
- **PDP:** average power multiplied by propagation delay; recalculated in Python rather than copied manually.
- **Energy per encoding operation:** average encoder power divided by the input-word rate used by the activity pattern.

- **Energy per encoded bit:** energy per operation divided by 11 or 12, consistently by codeword length.
- **Output swing:** maximum measured output minus minimum measured output.
- **Hardware complexity:** transistor count from the instantiated cell topology. It is not layout area.

Official Software Source

LTspice 26.0.2 was obtained from <https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>. The project uses .param, .temp, .tran, and .meas directives in retained circuit files.